CodeNeuraX

Online Hackathon On Cloud Technologies

Problem Statement: Create an Open Source lifecycle management tool for NIC including patch management
Description: The objective of this task is to create an Open Source complete lifecycle management tool for physical, virtual servers and Containers. Which would give system administrators the power and ability to easily automate repetitive tasks, quickly deploy applications, and proactively manage servers, on-premise or in the cloud.

Problem Statement: Open-source API testing tool
Description: The objective of this task is to find out best suitable tools for testing API functionality, reliability, performance, and security. JMeter was initially created for load testing purposes, but it was used for other purposes as well. These include security issues as well. Best open source API testing tools SoapUI,JMeter, Postman etc

Problem Statement: ModSec rules for WAF H/W or S/W Defined Appliances to application restrict attacks
Description: The objective of this task is to write ModSec rule sets for OWASP Top 10 and other custom web vulnerability patterns. These rulesets would be one of the major building blocks for the ModSec Based WAF solutions. These rule sets can be applied in any H/W or S/W WAF supporting ModSec rules.

Problem Statement: Create a Vulnerability Assessment Scanner
Description: The objective of this task is to create an Open Source vulnerability assessment scanner for application programs or networks. This tool should be integrated to cloud native orchestrators like OpenStack or Kubernetes. OWASP ZAP (short for Zed Attack Proxy) is an open-source web application security scanner. It is intended to be used by

both those new to application security as well as professional penetration testers.

Problem Statement: Open Source Backup Tool with control panel and GUI based control panel

Description: This solution should have the option to register clients using GUI/API based control panels. The solution should have options for monitoring and scheduling of backups and it's status. The team members can explore and use open source backed engines like Restic, Amanda, GoodSync, Duplicati, Backy2 etc.

Problem Statement: Unified monitoring solution for VMs and Containers

Description: Objective of this task is to create and integrate a unified monitoring solution for VMs as well as Container environments. The Solution should have options to scrape data from different required matrices and store it in a RTDB. There should be options to visualize the matrices and generate alerts on required options.

Problem Statement: Website Change Detection and Monitoring

Description: Web page change detection tools help you to track changes made on the websites. These tools enable you to find unplanned code changes.

Problem Statement: Self Service APIs or Connectors for Cloud Platforms viz. OpenStack, VMWare, Azure, AWS, Google, etc.

Description: The objective of this task is to create different cloud connectors which can be used to orchestrate tasks from Unified Self-Service cloud portals. All these connector modules can be written in Python or Go and are able to connect to REST API or RPC based solutions.

Problem Statement: Container Live Migration and BCP within and across the cloud platforms

Description: As the containers can hold microservices which can be shared with other applications running in the same POD or different POD

using services by grouping logical sets of PODs. This tight coupling makes the container availability critical. Live migration of containers helps moving applications between physical machines or clouds without disconnecting clients. This also helps in physical server maintenance, load balance of cluster, trouble shooting in cloud issues etc.